

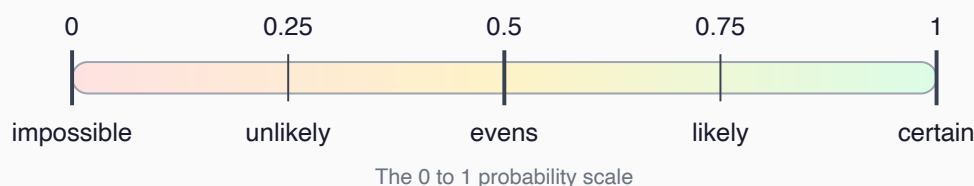
● RAG Learning

5.1 Probability

AQA GCSE Mathematics 8300 — Probability — spec refs P1–P9 — Papers 1 (non-calculator), 2 and 3 (calculator)

5.1.1 The Probability Scale and Basic Probability

Probability measures how likely an event is, on a scale from 0 (**impossible**) to 1 (**certain**). Probabilities can be written as **fractions, decimals or percentages** — but never as ratios or "1 in 4" in an exam answer.



Theoretical Probability

When all outcomes are **equally likely**:

$$P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}$$

Worked Example: A bag contains 3 red, 5 blue and 2 green counters. One is taken at random. Find $P(\text{blue})$ and $P(\text{not green})$.

$$\text{Total} = 10. \quad P(\text{blue}) = \frac{5}{10} = \frac{1}{2}. \quad P(\text{not green}) = \frac{8}{10} = \frac{4}{5}$$

Outcomes That Sum to 1 (P4)

The probabilities of an **exhaustive** set of **mutually exclusive** outcomes add up to 1. In particular:

$$P(\text{not } A) = 1 - P(A)$$

Worked Example: A biased spinner lands on red, blue, green or yellow. $P(\text{red}) = 0.3$, $P(\text{blue}) = 0.25$, $P(\text{green}) = 0.1$. Find $P(\text{yellow})$.

$$P(\text{yellow}) = 1 - (0.3 + 0.25 + 0.1) = 1 - 0.65 = \mathbf{0.35}$$

Worked Example: A four-colour spinner has $P(\text{red}) = x$, $P(\text{blue}) = 2x$, $P(\text{yellow}) = 3x$ and $P(\text{green}) = 0.1$. Find x .

$$x + 2x + 3x + 0.1 = 1 \Rightarrow 6x = 0.9 \Rightarrow x = \mathbf{0.15}$$

Exam Tip: Check every probability answer lies between 0 and 1. If you get 1.2 or -0.3 , you have made an error. Fractions do not have to be simplified unless the question asks, but must be correct.

5.1.2 Experiments, Relative Frequency and Expected Outcomes (P1, P2, P3, P5)

Relative frequency (experimental probability) = $\frac{\text{number of times the outcome occurs}}{\text{total number of trials}}$.

It is an **estimate** of the probability.

Results of experiments can be recorded in **frequency tables** and **frequency trees**. As the number of trials increases, the relative frequency tends towards the **theoretical probability** (P5). So a

larger number of trials gives a more **reliable** estimate.

Worked Example: A coin is thrown 200 times and lands heads 116 times. Estimate $P(\text{heads})$. Is the coin biased?

Relative frequency = $\frac{116}{200} = 0.58$. This is above 0.5, so the coin *may* be biased towards heads — but we cannot be certain. More throws would give a more reliable estimate.

Fairness and Randomness

An object is **fair** (unbiased) if every outcome is equally likely. Choosing “at random” means every item has the same chance of being picked. Even a fair dice will not give exactly equal frequencies in a small experiment.

Expected Outcomes (P2, P3)

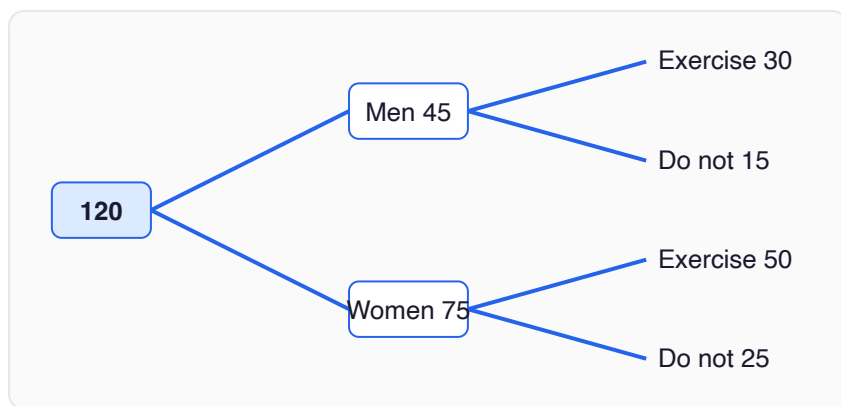
$$\text{Expected number of outcomes} = P(\text{outcome}) \times \text{number of trials}$$

Worked Example: $P(\text{win}) = 0.15$ for a game that costs £1 and pays £5 to a winner. It is played 200 times. Find the expected profit for the organiser.

Expected wins = $0.15 \times 200 = 30$. Takings = $200 \times £1 = £200$. Payouts = $30 \times £5 = £150$.
Expected profit = $£200 - £150 = \text{£}50$

Frequency Trees (P1)

A frequency tree splits a group into categories using **numbers**, not probabilities.



From the tree: $P(\text{a woman who exercises}) = \frac{50}{120} = \frac{5}{12}$, and $15 + 25 = 40$ people do not exercise.

5.1.3 Listing Outcomes and Sample Spaces (P6, P7)

A **sample space** (possibility space) is a list or table of every possible outcome. List outcomes **systematically** so none are missed — e.g. fix the first item and run through all choices of the second.

Worked Example: Ali chooses a starter (soup or salad) and a main (fish, pie or curry). List all combinations.

soup–fish, soup–pie, soup–curry, salad–fish, salad–pie, salad–curry ($2 \times 3 = 6$ combinations)

Sample Space Diagram for Two Dice

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9

4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

There are 36 equally likely outcomes. $P(\text{total} = 7) = \frac{6}{36} = \frac{1}{6}$ and $P(\text{total} \geq 10) = \frac{6}{36} = \frac{1}{6}$.

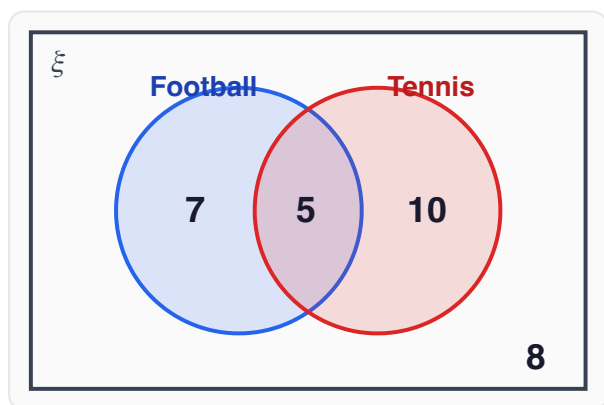
Worked Example: Two fair dice are rolled. Find the probability that the difference between the scores is 2.

Outcomes: (1, 3), (3, 1), (2, 4), (4, 2), (3, 5), (5, 3), (4, 6), (6, 4) — 8 of the 36.

$$P = \frac{8}{36} = \frac{2}{9}$$

5.1.4 Venn Diagrams (P6)

A Venn diagram shows how items are shared between sets. The rectangle contains everything being considered. **Always fill in the overlap first**, then work outwards.



Worked Example: 30 students: 12 play football, 15 play tennis, 5 play both.

Overlap = 5. Football only = $12 - 5 = 7$.

Tennis only = $15 - 5 = 10$.

Neither = $30 - (7 + 5 + 10) = 8$.

$$P(\text{tennis only}) = \frac{10}{30} = \frac{1}{3}$$

Set Notation Higher

Notation	Meaning	In the diagram above
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ξ	Universal set — everything	all 30 students
$A \cap B$	Intersection: in A and B	5
$A \cup B$	Union: in A or B (or both)	$7 + 5 + 10 = 22$
A'	Complement: not in A	not football: $10 + 8 = 18$
$A \cap B'$	In A but not B	football only: 7
$(A \cup B)'$	In neither	8

Worked Example: $\xi = \{1, 2, \dots, 12\}$, $A = \{\text{factors of } 12\}$, $B = \{\text{odd numbers}\}$. List $A \cap B$ and $A \cup B$.

$$A = \{1, 2, 3, 4, 6, 12\}, B = \{1, 3, 5, 7, 9, 11\}$$

$$A \cap B = \{1, 3\}, A \cup B = \{1, 2, 3, 4, 5, 6, 7, 9, 11, 12\}$$

5.1.5 Combined Events: When to Add and When to Multiply (P8)

Mutually exclusive events cannot happen at the same time. **Independent** events do not affect each other — the outcome of one does not change the probability of the other.

OR rule (mutually exclusive): $P(A \text{ or } B) = P(A) + P(B)$

AND rule (independent): $P(A \text{ and } B) = P(A) \times P(B)$

At least one: $P(\text{at least one}) = 1 - P(\text{none})$

Worked Example: $P(\text{Sam's bus is late}) = 0.3$ and $P(\text{his train is late}) = 0.2$, independently. Find the probability that (a) both are late (b) neither is late.

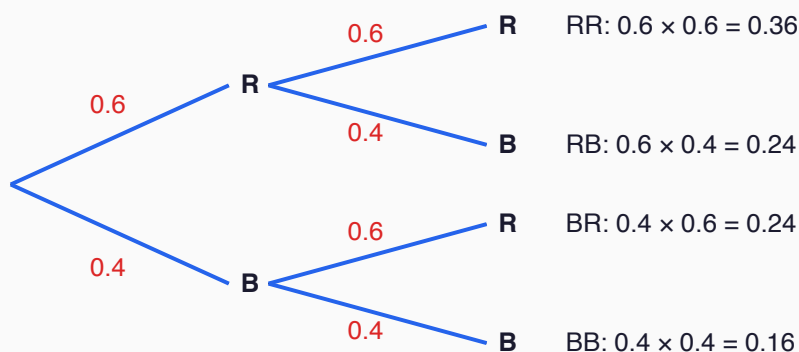
(a) $0.3 \times 0.2 = \mathbf{0.06}$ (b) $0.7 \times 0.8 = \mathbf{0.56}$

Worked Example: A fair dice is rolled twice. Find $P(\text{at least one six})$.

$$P(\text{no six}) = \frac{5}{6} \times \frac{5}{6} = \frac{25}{36}, \text{ so } P(\text{at least one six}) = 1 - \frac{25}{36} = \frac{11}{36}$$

5.1.6 Tree Diagrams (P6, P8)

Tree diagrams show two or more events in sequence. **Multiply along the branches** to find the probability of a combined outcome, and **add** the end results for "or". Each pair of branches from a point must add to 1.



A spinner with $P(\text{red}) = 0.6$ is spun twice. $P(\text{exactly one red}) = 0.24 + 0.24 = 0.48$. The four end results add to $0.36 + 0.24 + 0.24 + 0.16 = 1 \checkmark$

With and Without Replacement (Dependent Events)

If an item is **replaced**, the probabilities stay the same on each pick (independent events). If it is **not replaced**, the second probabilities change: the total goes down by 1, and so does the count of the colour already taken. These are **dependent** events.

Worked Example: A bag has 5 red and 3 blue sweets. Two are taken without replacement. Find $P(\text{one of each colour})$.

$$P(R, B) = \frac{5}{8} \times \frac{3}{7} = \frac{15}{56} \quad P(B, R) = \frac{3}{8} \times \frac{5}{7} = \frac{15}{56}$$

$$P(\text{one of each}) = \frac{30}{56} = \frac{15}{28}$$

Worked Example: A bag contains n sweets, 6 of them orange. Two are taken without replacement and $P(\text{both orange}) = \frac{1}{3}$. Find n .

$$\frac{6}{n} \times \frac{5}{n-1} = \frac{1}{3} \Rightarrow n(n-1) = 90 \Rightarrow n^2 - n - 90 = 0$$

$$(n-10)(n+9) = 0, \text{ so } n = 10 \text{ (reject } -9\text{)}.$$

Exam Tip: The assumptions behind your calculation matter: multiplying probabilities assumes the events are **independent**. In “without replacement” questions, look out for the words “takes two”, “eats” or “does not replace” — the second-pick denominator must drop by 1.

5.1.7 Conditional Probability (P9) Higher

Conditional probability is the probability of an event *given that* another event has happened, written $P(A \mid B)$. Knowing B has happened **restricts** the outcomes to those in B .

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

From Two-Way Tables

	Owns a bike	No bike	Total
Year 10	20	15	35

Year 11	30	15	45
Total	50	30	80

Worked Example: A student who owns a bike is chosen at random. Find the probability they are in Year 11.

Restrict to the "owns a bike" column: 50 students, of whom 30 are in Year 11.

$$P(\text{Year 11} \mid \text{bike}) = \frac{30}{50} = \frac{3}{5} \quad (\text{compare } P(\text{Year 11}) = \frac{45}{80})$$

From Tree Diagrams

Worked Example: $P(\text{rain}) = 0.3$. If it rains, $P(\text{Mia late}) = 0.4$; if not, $P(\text{Mia late}) = 0.1$. Given Mia is late, find the probability it rained.

$$P(\text{late}) = 0.3 \times 0.4 + 0.7 \times 0.1 = 0.12 + 0.07 = 0.19$$

$$P(\text{rain} \mid \text{late}) = \frac{0.12}{0.19} = \frac{12}{19} \approx 0.632$$

From Venn Diagrams

Worked Example: $P(A) = 0.45$, $P(B) = 0.3$, $P(A \cap B) = 0.1$. Find $P((A \cup B)')$ and $P(B \mid A)$.

$$P(A \cup B) = 0.45 + 0.3 - 0.1 = 0.65, \text{ so } P((A \cup B)') = \mathbf{0.35}$$

$$P(B \mid A) = \frac{0.1}{0.45} = \frac{2}{9}$$

Using Expected Frequencies

Converting probabilities to frequencies out of a convenient total (e.g. 100 or 1000 people) makes conditional questions easier: in the rain example, out of 100 days, 30 rainy days give 12 late days and 70 dry days give 7 late days, so 12 of the 19 late days were rainy.

Exam Tip: Look for the words “given that”, “of those who” or “a student who... is chosen”. The denominator is then the **size of the given group**, not the overall total. Independent events satisfy $P(A | B) = P(A)$.

Summary

Idea	Rule
Complement	$P(\text{not } A) = 1 - P(A)$
Relative frequency	frequency $\div$ number of trials
Expected frequency	probability $\times$ number of trials
Mutually exclusive (or)	$P(A) + P(B)$
Independent (and)	$P(A) \times P(B)$
At least one	$1 - P(\text{none})$
Conditional Higher	$P(A B) = \frac{P(A \cap B)}{P(B)}$